

Harness the Wind

Today we will learn, build, test, and show our thinking.

Grade 4 learners: big steps, clear checks, and one thing at a time.

Today's Goal

- ▶ Understand the concept of wind as a renewable energy source.
- ▶ Identify the parts of a windmill and their function.
- ▶ Explore how windmills convert energy into mechanical motion.

Remember

We do not have to be perfect right away. We take one step, then the next step, then we check.

What You Need

- ▶ Your class materials
- ▶ The digital tool or build materials for today
- ▶ A partner voice and your thinking

Partner Norms

Look. Listen. Try. Talk. Help without taking over.

Step 1

Look at the challenge and name the goal.

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- ▶ Point to the part that tells you what success means.
- ▶ Make the challenge criteria visible from the first slide through cleanup.

Pause. Check. Then move on.

Step 2

Build, test, and notice what happens.

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- ▶ Change one thing at a time so your next move is clear.
- ▶ Break the design process into plan, build, test, and improve rather than one big make.

Pause. Check. Then move on.

Step 3

Improve your design and explain your thinking.

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- ▶ Tell what worked, what did not yet work, and what you tried next.
- ▶ Use teacher talk that celebrates revision so early failure feels normal.

Pause. Check. Then move on.

If You Get Stuck

1. Read the directions on your screen or slide one more time.
2. Check the example, the checklist, or your partner role.
3. Try one small change before asking for a full rescue.
4. Students may rush into building before naming the target or the constraint.

Help Routine

Stop. Breathe. Point to where you are stuck. Ask a specific question.

Show Your Learning

- ▶ I stayed with the steps and used the help routine when I needed it.
- ▶ I made, coded, explained, or revised something that matches the lesson goal.
- ▶ I can point to one choice, fix, or idea I learned today.

Before You Leave

Save it, share it, explain it, or demonstrate it.

Teacher Voiceover Cues

- ▶ Today we are working on harness the wind. Watch first, then try it with me.
- ▶ If you feel stuck, stop at the checkpoint slide and use the help routine before you panic.
- ▶ Show what you learned by saving, sharing, explaining, or demonstrating one clear success move.

This slide can be removed before student use if you only want the visual deck.